

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month we visualise the world's ocean plastic waste, we cover the origins of hydroponics, we delve into a local version of The Cancer Genome Atlas, study the unknown genes in the body, white holes and spinal interfaces



Mosquitoes drawn to flowers too

For mosquitoes, the primary source of food is actually the nectar from flowers. Male mosquitoes feed exclusively on nectar, while female mosquitoes feed primarily on nectar. Mosquitoes use their sense of smell to hone in on flowers as well. <https://dgit.in/qitoflwr>

WHAT'S NEW

Khetworks designs low cost solar pumps for Indian farmers

Former MIT students have started deliveries of their solar powered water pump to farmers, through a startup known as Khetworks. Traditionally, the farmers depend on the monsoon season to grow crops, and were not able to cultivate their fields all year round. A major barrier was cheap and reliable access to electricity. During the non monsoon months, farmers moved to cities and work as labourers. While diesel based pumps could be hired, they were expensive and did not provide the farmers with a viable profit margin. Khetworks co-founder and CEO Katie Taylor initially began by designing drip



irrigation systems for farmers, but found a major problem was access to groundwater. Designs for the solar pumps by Khetworks were created from scratch

based on the requirements of farmers. Khetworks has set up a manufacturing unit in Pune, with the pump design being granted patents. The 320 watt solar powered water pump is being sold to farmers, internally referred to as "version one". They are benefiting entire villages already, since the roll out. <https://dgit.in/khetworks>

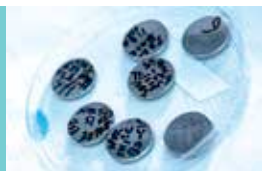
Hawking radiation may be confirmed by gravitational wave echoes

Stephen Hawking famously proposed that black holes, in fact, can radiate particles, because of quantum effects near the event horizon. This leak is known as Hawking Radiation. For the Hawking Radiation to exist, the region around the event horizon of newly created black holes must have a "quantum fuzz", regions of spontaneously created particle-antiparticle



pairs interacting with the event horizon. If the quantum fuzz does exist around the black hole, then they would reflect the gravitational waves created from say the merger of two black holes. Using gravitational wave

astronomy, scientists can detect the echoes of the gravitational waves, by detecting smaller signals that follow the gravitational waves of the collision event. For the first time, researchers from the University of Waterloo have published the first tentative findings of these gravitational wave echoes, which can confirm the existence of Hawking Radiation. <https://dgit.in/hawkrad>



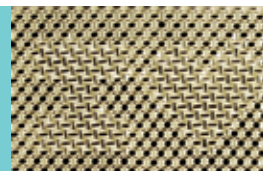
Better batteries

Researchers from Rice University have found out a way to increase longevity in lithium-ion batteries, by identifying a sweet spot in how much they should be charged. If the batteries are used without charging to their full capacities, they can be used for longer periods of time. <https://dgit.in/bettrbat>



Beauty sleep matters

Everyone and their grandmothers know that having a good night's sleep is essential for the next day to go well, now researchers from the University of Manchester have pinpointed exactly why. Researchers made the finding by genetically removing the "body clocks" of mice. <https://dgit.in/happynap>



Kirigami materials

Inspired by the Japanese art of paper cutting, researchers from Harvard have created programmatic kirigami materials that can transform their surfaces according to the need. Such materials could be used for sensors in robots and stretchable electronics. <https://dgit.in/kirmat>